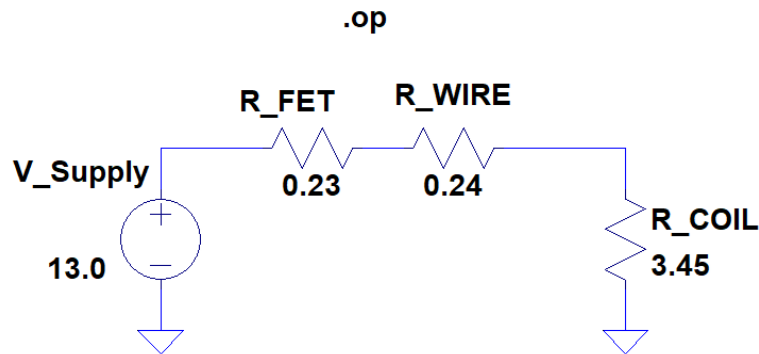


Polarization Coil Thermal Heating Considerations

The simple design of the polarizing circuit does not provide a constant current over time. Copper increases in resistance about 0.4% per degrees Celsius. The polarization coil pack can change from around 3.2Ω to around 3.7Ω . Due to coil heating, the polarizing current changes about 4% during the 33 phase scans.



--- Operating Point ---

V(R_coil):	11.4413	voltage
I(R_coil):	3.31633	current

Average coil heating was calculated from the power dissipation in the coil. Estimates of the coil thermal capacitance were obtained from the coil volume. An estimate of the thermal resistance coil to air was calculated using the coil surface area.

Thermocouple measurements of coil heating during operation aided in refining the thermal model values. Simulations were then performed using LTSpice and safe operating times were obtained.

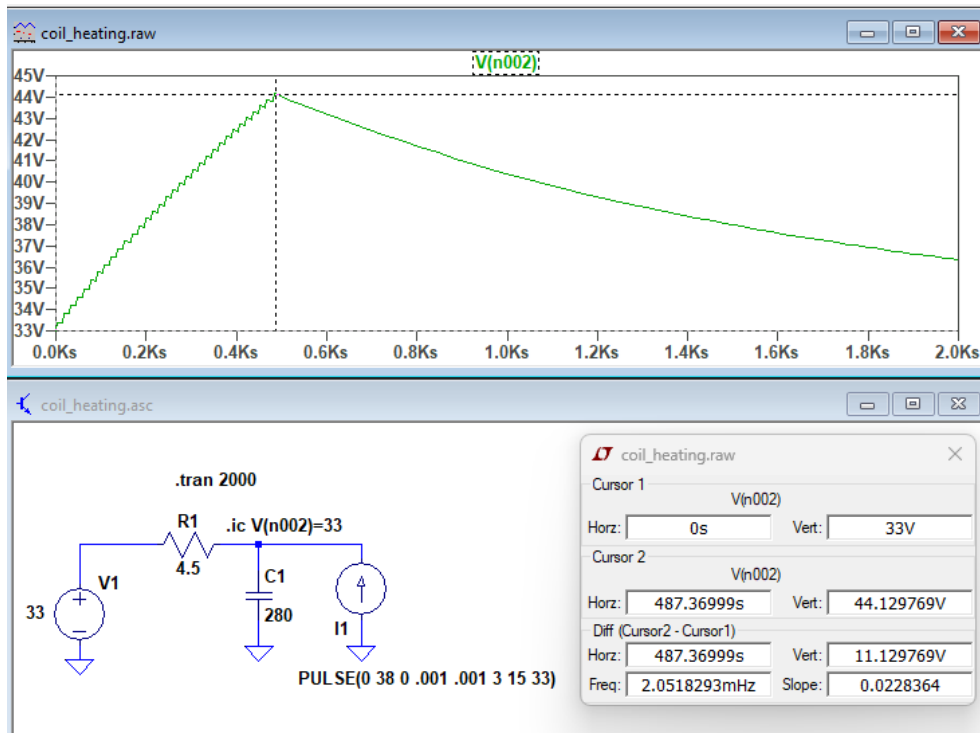
The electrical equivalents for the simulation are:

Heat Capacity J/°C = Capacitance in Farads

Thermal Resistance °C/Watt = Resistance in Ohms

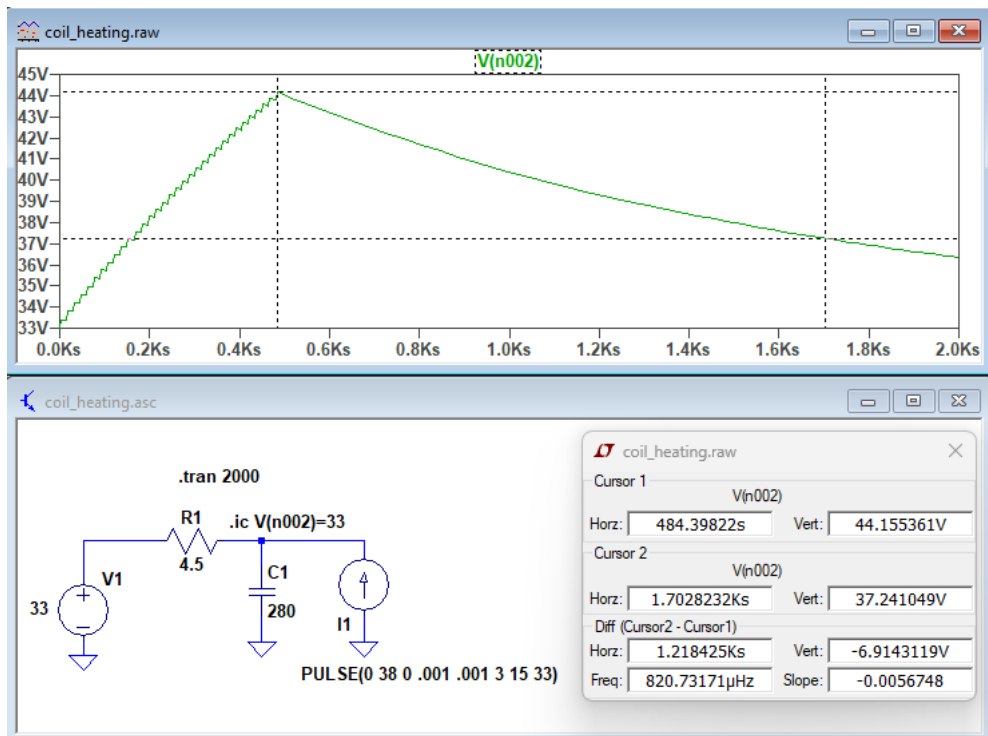
Heating in Watts = Current In amps

Temperature in °C = Voltage in Volts

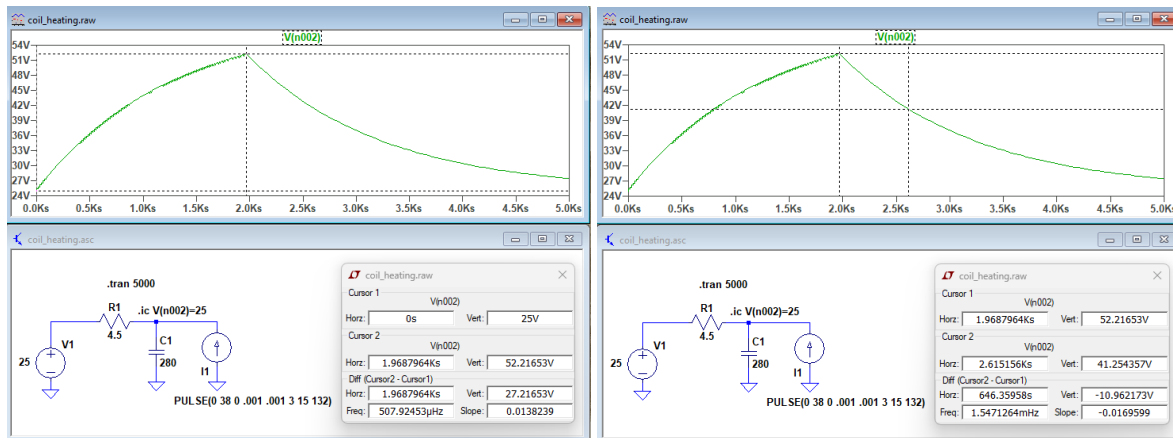


Coil temperature rise is about 11°C per full scan of 33 phases at 3 seconds polarization and 15 second intervals.

The model was derived from a test performed at an ambient temperature of 33°C. The PVC form is rated to around 60°C maximum. To keep the coil temperature no more than 60°C, limit operation to three consecutive full scans. Then, let the coil cool.



Coil cooling to the ambient temperature is slow. After one full scan, a 7°C drop took around 20 minutes. This drop was used to calculate the thermal resistance to air of the coil system.



Heating is not quite an exponential response due to the large heat capacity of the coil and the constant power input. Cooling is dependent only on the delta temperature. For example, if the ambient temperature is 25°C, the delta from 52°C is 27°C. It takes about 11 minutes to cool 11°C. The next full scan can be performed after this shorter wait.

In the simulation, four full scans in a row increased the temperature from 25°C to 52°C. At 25°C, it is recommended not to perform more than four sequential 33 phase full scans before allowing the coil to cool for at least 11 minutes between each successive full scan. Reduce the number of initial sequential scans when operating at higher ambient temperatures.

Cooling could also be accelerated with forced air or the use of an ice pack placed in the 2" tube.